

Solving quadratic equations



- 1
- a $y = 0, \frac{1}{5}$ b $x = -11, 5$ c $x = \frac{1 \pm \sqrt{161}}{8}$ d $m = 4 \pm \sqrt{10}$
- e $x = 9, -5$ f $x = -\frac{2}{3}, 5$

- 2
- a Yes b No c Yes d No

- 3
- a $x = -4, -5$ b $x = -4, -5$

Applications

4 $x = 3, -7$

- 5
- a 313.6 metres b $t = 4$

- 6
- a $t = 3$ b $t = 9$

7 $t = 6$

- 8
- a $m = 1900$ b 64.8 km/h c $v = 16$

- 9
- a $S = 528$ b $n = 5$

10 $n = 40$

- 11
- a $x = 7$ b 8

12 $x = 7, -8$

13

a $t = 2, 3$

b 5 sec



c No

14

a $t = \frac{1}{4}, 2\frac{3}{4}$

b $2\frac{1}{2}$ sec

c $t = 1\frac{1}{4}, 1\frac{3}{4}$

d $\frac{1}{2}$ sec

15

a $y = 14 - x$

b $x = 8, 6$

c $y = 6, 8$

16

a $w = 10$

b 16 metres

17 $t = 4$

18

a $x = 10$

b 23 metres

19

a $x = 9$

b 19

c 21

20 $x = 10$

21

a $r = 6$

b 6.6 m

22

a $4x^2 + 44x \text{ m}^2$

b $x = 2$

23

a $m = 10$

b 15 hours

24

a $w = 2.33$

b 15.3 cm

25

a 41

b $70 - x$

c $70x - x^2$

d 1225

e Yes

26

a $x = 5, -1$

b $(5, 7), (-1, 7)$



27 $x = 8$



28

a $x^2 + 7x \text{ m}^2$

c $x = 23$

b $x^2 + 23x + 120 \text{ m}^2$

d Width = 23 m, Length = 30 m

Further quadratic equations

29 $x = 0, 3$

30

a $\frac{x^2 + y^2}{xy}$

b $xy = 12$

c $x = -4, -3, 3, 4$

d $y = -3, -4, 4, 3$

e $\frac{3}{4}$

31

a C

b $x = \frac{-1 + \sqrt{5}}{2}$

32 $x = \frac{1 + \sqrt{33}}{2}$

33

a $x = -1, -4$

b $x = 2, -3, -5$